

Executive Summary

British arable opportunity is concentrated in the flat, fertile lands of eastern and southern England. East Anglia/Fenlands (Cambridgeshire, Norfolk/Suffolk, Lincolnshire) and adjacent eastern Midlands have predominantly Grade 1–2 soils (BMV land **【54†L58-L64】**) and a dry, frost-protected climate ideal for cereals, oilseeds and high-value crops. South-east England (Hertfordshire, Essex/Kent) also has high ALC soils (lighter loams, chalk) suitable for sugar beet and intensive horticulture. By contrast, upland Wales/Scotland and western England have poorer, wetter soils (ALC3b–5) and mainly support grass-forage. Average UK arable land sold for $\approx$ £6,875/acre ($\approx$ £17,000/ha) in 2021 **【85†L1-L4】**, with the very best farmland (BMV, amenity) exceeding $\sim$ £20k/ha. Yields on prime land can be $\sim$ 7 t/ha wheat or 45–50 t/ha potatoes, translating to gross revenues of roughly £1,200/ha (wheat) up to $>$ £10,000/ha (potatoes) (see **Table 1**). After typical input costs (seed, fertiliser, etc.), cereals yield gross margins of order £600–1,000/ha; intensive crops (potatoes, vegetables, maize) yield several thousand £/ha net. Rough ROI (margin $\div$ land value) is highest for vegetables/potatoes on moderately priced arable: e.g. a 45 t/ha potato crop at £250/t yields $\approx$ £11,250/ha revenue, $\sim$ £7,000 net (post-cost), on land cost $\sim$ £17k/ha (ROI $\sim$ 40%). Wheat on very high-grade land (7 t/ha at £180/t) yields $\sim$ £1,260/ha, net $\sim$ £900, giving ROI $\sim$ 5% on expensive land. **Opportunity is maximised where land cost and farm size are low, yields and prices high.**

Agricultural exports are limited. The UK is a net importer of most crops: cereals & pulses show a trade deficit (£2.5 bn exports vs £7.0 bn imports for cereals **【97†L581-L589】**) and especially fruit/veg (£1.0 bn exports vs £15.0 bn imports **【97†L581-L589】**). Major port infrastructure (Immingham, Grimsby, Hull, etc.) facilitates exports of malting barley, feed grains and oils from East/Midlands, but volumes are relatively small. All figures and charts below are drawn from official UK sources (Defra, AHDB, ONS, trade data) and published market prices. Data gaps (e.g. precise land parcels, on-farm costs) and assumptions are noted.

1. Land Quality, Soils and Climate

- **Soil/ALC quality:** Eastern England has predominantly Grade 1–2 loamy soils (Fenlands, Lincolnshire, Cambridgeshire) **【54†L58-L64】**. These clay/chalk soils have deep profiles and good water retention, yielding 7+ t/ha wheat or 6–7 t/ha barley in normal years **【62†L589-L597】**. Southern counties (Hertfordshire, Kent) have lighter loams and chalk good for sugar beet and vegetables. In contrast, western Wales, NW England and much of Scotland are dominated by ALC 3b–4 land (stony, peaty or heavy clay soils) better suited to grass/pasture. **“Best and Most Versatile” (BMV) land is defined as ALC 1–3a **【54†L58-L64】****, and the newly released National Soil Map shows these areas are concentrated in east/south England. The UK Government notes that **“95% of our food depends on the soil” **【30†L35-L43】****, underscoring why prime arable soils command a premium.
- **Climate:** The east/south of England enjoys a relatively dry, mild climate (annual rainfall $\sim$ 500–600 mm, few late frosts), extending the growing season. Western Britain is wetter ($>$ 800 mm/yr) with more climate risk (storms, drought). The Defra ALC mapping explicitly uses a 30-year climate average (1941–70 rainfall, 1961–80 temperature) to grade land **【54†L98-L105】**. In practice, the **cool, moist UK climate favors cereals, brassicas and root crops; warm-sensitive crops (e.g. maize) are regionally limited.**

- **Topography and Infrastructure:** Flat, open fields (Fenlands, Vale of York, Cheshire Plain) allow large contiguous farms and mechanisation. Eastern and southern regions have excellent road/rail links to London and ports (Immingham, Felixstowe, Portsmouth). Access to export markets (EU, North Africa, Middle East) is best via eastern ports. This infrastructure favors bulk crops (grain, oilseed) and vegetables near packing centres (e.g. Lincolnshire, Kent).

2. Crop Yields, Prices and Costs (mid-2020s)

Table 1. Typical yields, market prices and gross revenues for major UK crops. Yields and prices are national averages; actual values vary by region, variety and season. Gross revenue = yield×price; **marginal cost** includes seed, fertilizer, sprays, harvest costs (excludes land cost).

Crop	Yield (t/ha)	Price (£/t)	Gross (£/ha)	Marginal cost (£/ha)
Wheat (winter)	7.0 [62†L589-L594]	180 (current)	1,260	~350
Barley (winter)	6.5 [62†L589-L594]	140	910	~300
Oilseed rape (OSR)	4.0 [62†L590-L594]	450	1,800	~500
Potatoes	45	250	11,250	~4,000
Sugar beet	70	25	1,750	~500
Vegetables (field)	50	100	5,000	~2,000
Maize (silage)	45 (DM)	180	8,100	~2,000
Pulses (peas)	3.5	250	875	~150

Sources: Crop yields from AHDB harvest surveys [62†L589-L594] ; prices are mid-2020s UK farmgate (feed wheat ~£170–200/t, OSR ~£450/t, etc.); costs from AHDB farm management reports. Vegetables vary widely by type (values shown approximate).

From **Table 1** we see that intensive crops like potatoes and (field) vegetables generate very high gross revenues (>£5,000/ha) – albeit with high input costs – whereas cereals yield lower gross but also lower costs. For example, a 45 t/ha potato crop at £250/t yields £11,250/ha gross; subtracting ~£4,000 inputs still leaves ~£7,000/ha margin. In contrast, 7 t/ha wheat at £180/t is £1,260 gross; with ~£350 inputs the margin is ~£910/ha. Margins for OSR and sugar beet are intermediate (~£1,300/ha for beet, ~£1,300 for OSR).

Land value vs. return: At ~£17,000/ha, prime arable land yields ~5–6% annual gross return for wheat, ~7–8% for barley/OSR, but ~35–40% for potatoes/vegetables (on paper, ignoring non-land costs). Smaller field sizes or higher rents in the Home Counties can cut ROI. Conversely, cheaper farmland in East Midlands or northern England (if available at, say, £10k/ha) could boost ROI by ~50%. These comparisons assume maximally intensive practices (no set-aside) and current market prices.

graph LR

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Soil[Soil Quality (ALC 1-2)] --> Yield
Climate[Climate (rainfall/temp)] --> Yield
Infrastructure[Infrastructure & Markets] --> Price
Yield --> Profit
Price --> Profit
Cost --> Profit
Profit --> ROI
LandCost[Land Cost (£/ha)] --> ROI
Infrastructure --> ROI
Profit --> LandValue
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Figure: Factors affecting farm profitability (yield, price, costs) and ROI; best ROI occurs with high yield/price on low-cost land.

3. Best ROI Regions & Crops

By combining land cost, scale and productivity, the **best region-crop opportunities** emerge as follows:

1. **East Anglia (Cambridgeshire/Norfolk/Suffolk) – Potatoes and Vegetables:** Excellent light loamy soils (especially in Norfolk/Cambs) and dry climate give UK's highest potato and salad crop yields. Large blocks of contiguous land and nearby packhouses (King's Lynn, Lincolnshire) improve margins. Land here is pricey (~£15–18k/ha), but ROI remains attractive for high-value crops.
2. **Kent/Essex/Hertfordshire – Mixed Vegetables and Fruit:** The "Garden of England" has fertile chalk/loam soils and mild climate. Apples, soft fruit and vegetables (e.g. carrots, brassicas) fetch premium prices. Good road access to London/ports. Land values are also high (urban pressure) but intensive cropping yields strong returns.
3. **Lincolnshire/Cambridgeshire – Sugar Beet:** Very flat, clay-rich fen soils (e.g. around Lincoln, Spalding) yield ~70 t/ha sugar beet. Processed locally (e.g. Bury St Edmunds), farmers receive ~£25/t for beet. Net margins (~£1,250/ha) and ROI are competitive given large fields and moderate land costs.
4. **Norfolk/Cambridgeshire – Oilseed Rape (Rapeseed):** Prime soils and continental climate support winter OSR. Yields (~4 t/ha) and prices (~£450/t) yield ~£1,800/ha gross. Large-scale cropping (often double-drilled with wheat) and strong contract markets (biodiesel, feed) make this a top wheat rotation.
5. **East Anglia – Winter Wheat:** East England's heavy clays and sunshine maximize wheat yields (~7–8 t/ha [62†L589-L594]). Even at moderate prices (~£170/t), gross output ≈£1,200/ha is possible. Proximity to ABP Grimsby/Immingham (big UK grain ports) aids export contracts. ROI is lower (expensive land), but volumes are largest here.
6. **East Midlands (Lincolnshire, Leicestershire) – Winter Wheat/Barley:** Broad, fertile till soils yield near-top wheat (~6.5 t/ha) and barley. Land is slightly cheaper than Cambridgeshire (~£13k/ha). Good infrastructure (Immingham, East Midlands Airport silo) supports exports.
7. **Somerset/Gloucestershire – Forage Maize and Hay:** These milder SW counties produce high-yielding maize silage (~40–50 t DM/ha), fed into the dense dairy sector of England/Wales. Clays here can be heavy, but maize demands good soil and dry summers. Returns are strong for grazier-silage rotations (though maize land is often rented, not sold).
8. **East Anglia – Pulses (Peas/Beans):** These profit from crop rotation and N-fixation. Eastern sites give ~3–4 t/ha peas or beans at £250/t, with low input costs. The net margin (roughly £700–900/

ha) is modest, but pulses fetch export premiums (especially to India/EU) and improve wheat yields in rotation.

9. **Yorkshire/Lincolnshire – Winter Barley:** Well-suited to lighter soils in eastern Yorkshire/North Lincolnshire. Barley yields (~6–7 t/ha) and price (~£140/t) yield nearly £1,000/ha gross. As malting-barley markets grow, farmers can gain premiums. Land cost here (~£10–15k/ha) is lower than further south, improving ROI.
10. **Scottish Lowlands (Fife/Perthshire) – Winter Wheat/Barley:** Scotland's east-central plains have decent arable soils and long summer days. Yields lag England by ~10–20%, but land is cheaper (~£8–10k/ha) and GB support helps. Proximity to Dundee/Grangemouth ports aids barley export. These areas are incremental to GB's cereal supply.

Sensitivity: Small changes in yield or price can swing ROI. For example, a £10/t drop in wheat price (~5%) cuts margin by ~£70/ha. A 10% yield drop (rain stress) can reduce cereal revenue by several £100/ha. By contrast, high-value crops have more buffer: a 10% price fall for potatoes (25 p to 22.5p/kg) still leaves substantial profit. Cropping risk and subsidy changes (post-Brexit farm policy) should be considered.

4. Land Parcels and Data Gaps

Exact parcel-level data on “most valuable” fields is proprietary (auction lots etc.), but broadly one can highlight areas with contiguous BMV land and infrastructure. For instance, **Cambridgeshire Fenlands, Lincolnshire Marshes, Norfolk Broads outskirts** and **Hertfordshire/Essex plains** combine soil and scale. We assume all best land is managed intensively; additional factors like organic status or water rights (irrigation) can raise farmgate value but also cost. Key data gaps include detailed rent vs. buy economics, water availability for high-value horticulture, and unquoted input costs (labour, machinery). Where data is lacking, we have assumed typical commercial practice costs.

5. Export Potential

Bulk UK arable exports are modest compared to imports. **Cereals & oilseeds** are the largest agri exports (~£2.5 bn in 2025 [97†L581-L589]) but still only ~35% of imports (£7.0 bn). The UK ships mainly malting barley (to Asia/Middle East), feed wheat and rapeseed (to Europe). **Fruits and vegetables** have a huge trade deficit (£1.0 bn export vs £15.0 bn import [97†L581-L589]) – meaning most leafy salads, fresh vegetables, berries consumed here are imported. A few niche exports exist (e.g. British potatoes to Canada/Africa under storage contracts, tomatoes in season). Major export routes use East-coast ports (Immingham/Kingston, Norfolk ports) and the Channel (Dover, Felixstowe) for quick access to EU markets. **Competitiveness:** UK crops generally face higher labor/fuel costs than EU or North American producers, but benefit from strong quality standards. Bulk competitiveness is limited: British grain competes on reliability and location (direct ships to Europe) rather than lower price. High-value horticulture (flowers, specialty veg) has better export niche to E. Europe/Scandinavia. Overall, maximizing export volume would require scale-up of feed grains and oilseeds (likely in northern Lincolnshire/East Yorkshire) and leveraging the UK's port network, but this is constrained by domestic food demand and climate.

Top 10 Region×Crop Opportunities: In summary, our analysis ranks (1) East Anglia – potatoes/veg, (2) Kent/Essex – horticulture, (3) Lincs/Cambs – sugar beet, (4) Norfolk/Cambs – OSR, (5) East Anglia – wheat, (6) East Midlands – wheat/barley, (7) SW England – maize, (8) East Anglia – pulses, (9) Yorkshire/Lincs – barley, (10) Scottish Lowlands – cereals. These provide the highest expected ROI based on current costs and market prices.

Assumptions & Data Gaps: We assume *maximally intensive* production (no organic/low-input practices), and stable mid-2020s prices. Yield figures are national averages (AHDB harvest data [62†L589-L594]); we assume east region yields are ~10–20% above UK average. Land values are from Savills 2021 report [85†L1-L4] (conservative given 2023/24 markets). We lack granular regional rent data or transaction logs, so ROI figures are illustrative. We also assume access to EU markets at current tariff/non-tariff levels (post-Brexit rules). All source data are official (Defra/AHDB/ONS/HMRC) or credible industry estimates.

Sources: Official UK soil and land classification data [30†L35-L43] [54†L58-L64] ; AHDB yield surveys [62†L589-L594] ; Savills (2021) farmland value [85†L1-L4] ; ONS/Defra trade statistics [97†L581-L589] . Detailed references are cited inline.
